

Supply-side arm: identifying eta_s in the BETA closure

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Version 4 (September 2026) **Version 3** (September 2026) **Version 2** (September 2026) **Version 1** (September 2026)

This document is the workplan for the supply-side arm, the cheapest of the candidates that restore variety inside the {BF, ALPHA, BETA} block (section 4.2 of docs/DOCS_ASSESSMENT.md, ranked candidate 2). House rule: plans normally live in ROADMAP.md or in an ADR; this file exists at your request, and the durable records remain the design file, the registry/scenarios.csv rows, the run manifests and the ADR that authorises the run.jl change. The short answer to the question that prompted it: **yes, under a supply shock BETA separates from ALPHA and from itself** — the row stops being a duplicate and becomes an identification result. The pilot already exists and its numbers are quoted below.

1 Why the demand-only matrix cannot see eta_s

The reported real-GDP metric is the household-consumption Tornqvist index, and under a demand-only shock the price block is pinned by technology alone: the zero-profit conditions $p_i = c_i(p, w)$ are homogeneous of degree 1 in (p, w) and contain no demand term, so they determine only the ratio $p/w = p^*(A)$. With the CPI numeraire, $w = 1/\text{CPI}(p^*)$, and at the calibrated technology ($A = 1$, hence $p^* = 1$) that gives $w = 1$ and $p = 1$ for **any** demand vector. The BETA supply curve $L^s = \bar{L} [(w/P)/(w_0/P_0)]^{\eta_s}$ then returns $\bar{L}$ for every η_s , which is why BETA reproduces ALPHA bit-for-bit (measured: max abs difference in quantities 1.1×10^{-16} , in prices 0).

Two corollaries that the session measured and that bound the design:

- A demand shock of ten times the programme size still gives $w = 1.0000000000$ and $\max |p - 1| = 2.66 \times 10^{-15}$; no demand-side re-specification can move the real wage.
- The numeraire is not the lever either (DE-0011): re-gauging changes the level of (p, w) , not their ratio.

The only channel that moves w/P is technology, i.e. `shocks.supply_shock`.

2 The pilot that already exists

experiments/probes/probe2_identification_and_shocks.jl, section D: a +20 % productivity shock in sector 1 ($A_1 = 1.2$, all other sectors $A = 1$), solved at $\theta = \epsilon = 0.5$, $\sigma = 0.9$, $\eta = 1$, warm-started from the stored ALPHA solution. Residual below 5.4×10^{-13} in every cell.

row	w (real wage)	L	real GDP F1	real GDP F2	real GDP F3
ALPHA ($\eta_s = 0$)	1.0049029745	1.0000000000	1.0055526180	0.9881460294	1.0051598783
BETA $\eta_s = 0.5$	1.0049029745	1.0024484897	1.0086846455	0.9912768336	1.0082906826
BETA $\eta_s = 2$	1.0049029745	1.0098299882	1.0181268156	1.0007153159	1.0177291649
BETA $\eta_s = 5$	1.0049029745	1.0247564458	1.0372202492	1.0198012921	1.0368151411

Three things follow, and they are the substance of the answer:

1. **BETA moves, ALPHA does not.** Employment is pinned at exactly 1 for ALPHA and rises with η_s for BETA: +0.24 %, +0.98 %, +2.48 % at $\eta_s = 0.5, 2, 5$. Under F1 the headline moves from +0.56 % (ALPHA) to +0.87 / +1.81 / +3.72 % — a spread of more than three percentage points across the elasticity grid, against a spread of 1×10^{-16} in the demand-only matrix.

2. **The elasticity is recovered exactly.** $\ln L / \ln w$ returns 0.500000, 2.000000, 5.000000; the pilot is therefore also a specification test of the implementation.
3. **Prices move at last** ($\max |p - 1| = 0.196$), which is the precondition for the other candidates (markups, capacity) to have anything to bite on.

3 Status of the arm [v3, 2026-09-20]

Version 3 records three things: the schema ADR is drafted (ADR-0023, proposed); the bridge step is done — the pilot is re-measured on the sectoral kernel by **probe15** (2026-09-19), whose numbers supersede the table above as the current evidence; and an identification-vehicle decision is fixed. The arm still changes the experiment, not **src/**: no closure, no kernel, no generation.

The bridge measurements (probe15, full-71, $A_1 = 1.2$; recorded in docs/log/2026-09.md). The η_s rungs separate: employment 1.000000 / 1.002187 / 1.008218 at $\eta_s = 0 / 0.5 / 2$ (sectoral uniform vector, F2), against a 0.02 % employment movement over the same ladder under the demand-only programme; the real-wage spread reads 1.215 / 1.105 / 1.043. The rigid-corner nesting (Proposition 3) is *shock-independent*: the canary is bit-exact with and without the shock. Two cautionary measurements: the shock alone moves prices ($\max |p - 1| = 0.226$ with no programme), so under a supply shock $\max |p - 1|$ is not a demand-sensitivity measure — the demand signature remains the F1 vs F2 = F3 difference; and the welfare effect flips sign across the rows (supply-only +1.20 %, supply + programme -0.40 %, demand-only -1.57 %), i.e. the two channels do not add linearly.

Identification vehicles (the v3 decision). The primary identification claim runs on the **scalar single-wage BETA** cells (2N+2, the executed matrix form), because the preregistered signature is scalar: $\ln L / \ln(w/P)$ recovers the input η_s to 10^{-6} in every cell and is invariant to the shock magnitude within a shock id. Each design additionally carries the **sectoral uniform vector** cells (3N+1, $\eta_{s,i} = \eta_s$ for every i) as the robustness row: there the identification is a vector ($\eta_{s,i} = \ln(L_i/\bar{L}_i) / \ln(w_i/\Pi)$ per sector) and no scalar signature applies; report per-sector recovery, never one number. Both forms are live in the kernel and in the executed v9 design (matrix cells = scalar, ladder cells = sectoral), so no promotion is needed. The ADR-0023 schema records this and keeps the two rows separate in every table.

4 The executed arm [v4, 2026-09-20]

supply_etas_s1 is executed: 63/63 cells, all gates pass (design commit 202837a; ADR-0023 schema; flow table paper/tables/supply_etas_flows.md, generated from the manifests). The ratified matrix: ALPHA control + scalar single-wage BETA ($\eta_s \in \{0.5, 1, 2, 5\}$, the identification row) + sectoral uniform BETA ($\eta_s \in \{0.5, 2\}$, the robustness row), times F1/F2/F3 times $A_1 \in \{1.1, 1.2, 1.3\}$.

The preregistered signatures all hold. (i) **Identification exact and magnitude-invariant:** the implied elasticity $\ln L / \ln w$ recovers the input η_s to four decimals (0.5000 / 1.0000 / 2.0000 / 5.0000 at $A_1 = 1.2$, F2) and its spread across the magnitude ladder is below 10^{-6} . (ii) **ALPHA is the control:** $L = 1$ exactly, $w = 1.0049 > 1$. (iii) Real GDP is monotone in η_s within each financing column. (iv) **Financing neutrality extends to supply shocks:** F2 = F3 in every metric; F1 distinct (the programme tilt). (v) **The canary under $A \neq 1$:** the ADR-0019 external-account identity is asserted on all 63 cells (all pass; **canary_diff** at solver noise) — the ETAs gate that had only ever been tested at $p = 1$ is now recorded under a technology shock. Prices move through technology ($\max |p - 1| = 0.196$ at $A_1 = 1.2$ for the scalar rows; the sectoral 3N+1 rows carry their own wage dispersion, $w = 1.007$, $\max |p - 1| = 0.228$ at $\eta_s = 0.5$).

Remaining: **supply_etas_prog** (productivity gain $A_i = 1 + \alpha\psi_i$ on the programme's own sectors, $\alpha \in \{0.05, 0.10, 0.20\}$) and **supply_etas_unif** (uniform A_i), same schema and batch pattern. The DOCS_ASSESSMENT section 4.2 write-up is recorded for the manuscript pass (the assessment document's revision palette has no colour beyond **revisionV6** yet).

5 Design of the arm

5.1 Shock specification

Three candidate shocks, in ascending cost. The recommendation is to run all three, because they answer different questions.

id	shock	what it identifies	note
s1	$A_1 = 1.2$, single sector (agriculture)	the pure identification test	already piloted; surgical, but economically arbitrary

id	shock	what it identifies	note
prog	$A_i = 1 + \alpha\psi_i$ on the programme's own sectors ($\alpha \in \{0.05, 0.10, 0.20\}$)	the economically relevant case: a green-investment programme bids up the wages of the sectors it buys from	the impulse is 63 % specialised construction; this shock targets exactly the margin the manuscript argues about
unif	$A_i = 1.01$ for all i	the cleanest real-wage experiment; uniform, so no composition contaminates the estimate	small magnitude: a uniform 20 % shock would move the real wage by 25 %

5.2 Cells and run ids

Design `supply_etas` in `experiments/designs/supply_etas.toml`; ids follow `<design>-<labor>-<financing>[-<variant>]` (ADR-0002). One design per shock id keeps the manifests readable: `supply_etas_s1`, `supply_etas_prog`, `supply_etas_unif`.

- Labour rows: ALPHA (the $\eta_s = 0$ control) and BETA at $\eta_s \in \{0.5, 1, 2, 5\}$, e.g. `supply_etas_s1-BETA-F2-etas2`.
- Financing: F1, F2, F3 as in the matrix, so the arm nests the published results.
- Shock ladder: three magnitudes per shock id, so the recovered elasticity can be checked for invariance.
- Count: $5 \times 3 \times 3 = 45$ cells per shock id; the pilot suggests a few seconds per cell, so each design is a single batch.

5.3 What “estimating η_s ” means here

Inside the model η_s is a parameter, not an estimate: at a solved cell it is recovered exactly by inversion,

$$\eta_s = \frac{\ln L}{\ln(w/P)},$$

and the pilot returns the input values to six decimals. The arm therefore delivers two publishable quantities rather than a point estimate:

1. **The identification result:** the map $\eta_s \mapsto (w/P, L, \text{real GDP})$ is strictly monotone and invertible under a supply shock, and degenerate (constant) under a demand-only shock. That is the claim the referee demand R2 actually needs.
2. **The sensitivity band:** the semi-elasticity of the headline to η_s , reported as the spread over $\eta_s \in [0.5, 5]$ per financing closure — i.e. how much of the published result rests on an elasticity the data do not pin down.

An empirical η_s (from an observed $\Delta \ln L$ against $\Delta \ln(w/P)$ pair) then enters the model as a scenario, not as a calibration; the arm is what makes that step informative.

5.4 Preregistered signatures

Recorded with the design (ADR-0006), deviations are findings (ADR-0004):

- $\ln L / \ln(w/P)$ reproduces the input η_s to 1×10^{-6} in every BETA cell, and is invariant to the shock magnitude within a shock id.
- ALPHA: $L = 1$ exactly, $w > 1$; BETA: L strictly increasing in η_s .
- Monotone ordering of the real-GDP headline in η_s within each financing column.
- Residual gate 10^{-6} (mobile); budget gate 10^{-9} .

5.5 Gates and open items to check, not to assume

- **Canary under $A \neq 1$.** The ADR-0010 identity $p \cdot \text{mktc} = S - (I + X - M) + T$ has only ever been asserted at $p = 1$. Zero profit still holds, so the derivation should carry, but the terms are now valued at moving prices: assert it on the first supply cells as a diagnostic and record the result (an ADR if it fails). **Status (v3): partly measured, not assumed — probe15 row set 3 asserts the rigid-corner nesting (Proposition 3) bit-exact with and without $A_1 = 1.2$, and the external-account identity is asserted on the first arm cells as planned.**

- **Scale determinacy.** The round-gain criterion $\max(A_{\text{bill}}/\lambda + (1 - m)(1 - s)fs) < 1$ uses the baseline intermediate coefficients; a productivity shock changes the optimal input mix at $\epsilon = 0.5$, so the criterion must be re-evaluated for the arm.
- **BF in the arm.** The recombination corner ($\eta = 0$ with $\eta_s > 0$) becomes informative only here, but it inherits the identification defect of ADR-0017: report its aggregates, never its quantities, until sector-specific wages land.

6 Workplan

1. **ADR-0018 (blocking).** `experiments/run.jl` hard-codes `Shocks(ones(N), ones(N), zeros(N))` in `build_cell_model` and `solve_cell`; the arm needs a per-cell supply-shock source and magnitude from the design file. The ADR fixes the schema, the scenario-id convention for η_s variants, and the rule that the baseline reference stays the no-shock one (real GDP remains “against the baseline”). *Status (v3): the schema ADR is drafted as ADR-0023 (proposed, 2026-09-20) — per-design [shock] block, default kind = "none" (existing designs bit-identical), the ADR-0002 id convention with the shock id in the variant slot, and the no-shock reference unchanged.*
2. **Bridge, before the ADR lands.** Extend the pilot to the full grid as `experiments/probes/probe5_supply_arm.jl` (read-only, no `src/` change), so the numbers exist and the signatures can be preregistered against measurement rather than expectation.
3. **Registry.** Append planned rows to `registry/scenarios.csv` for every cell (45 per shock id), with the parameter cells, the shock id and magnitude, and `eta_s`.

4. **Design and preregistration.** Write `experiments/designs/supply_etas_<id>.toml`, then

```
julia --project=. experiments/run.jl --list supply_etas_s1
julia --project=. experiments/run.jl --preregister supply_etas_s1 --actor hermes-agent
```

and commit the preregistration record before any cell is created.

5. **Execution.**

```
julia --project=. experiments/run.jl --design supply_etas_s1 --actor hermes-agent
```

then `supply_etas_prog` and `supply_etas_unif`.

6. **Gates after every batch.**

```
julia --project=. scripts/status.jl          # 0 warnings
julia --project=. scripts/check_repo.jl      # 0 violations
julia --project=. -e 'using Pkg; Pkg.test()'
```

7. **Write-up.** A paper table citing the `supply_etas-*` run ids: rows ALPHA and BETA (η_s grid), columns F1/F2/F3, cells = real GDP and employment, plus the recovered elasticity column. Update `docs/DOCS_ASSESSMENT.md` (section 4.2, the ranked proposal) and `registry/closures.toml` [`labor.BETA`] open gates; one `docs/log/2026-09.md` entry.

7 What the arm does not do [extended v2]

The arm moves prices **through technology only**: the zero-profit block contains no demand term, so $p \neq 1$ requires $A \neq 1$. Put the demand-only programme back in and the matrix returns to $p = 1$, $w = 1$, $L = \bar{L}$ — BETA collapses onto ALPHA again. The supply-side arm is therefore a **different experiment**, not a repair of the published 5 x 3 matrix: it identifies η_s and it prices the elasticity, but it does not restore variety inside the demand-shock block.

Variety *in the matrix the paper publishes* requires the price block to depend on demand, i.e. one of:

- **demand-sensitive factor prices** — sector-specific wages (ranked candidate 1): N sectoral labour markets make w_i , and through zero profit p_i , functions of sectoral labour demand. Zero profit is preserved (no rents to dispose of), it repairs the ADR-0017 defect, and it is the only variant that is *earned* by the model rather than assumed: the new parameters are sectoral supply elasticities of the same family as η_s .
- **a demand-sensitive markup** $\mu_i = 1 + \kappa(y_i/y_{i0} - 1)$: nested at $\kappa = 0$ (the current model), a few lines of code, and a clean sensitivity ladder. But markup income is profit, so it needs a profit-income-and-spending closure (who

receives the rent, and how it is spent) or the accounting identity behind the canary breaks — that is the real cost, not the code.

- **a fixed factor with capacity:** the standard short-run mechanism, but the largest surgery — a second factor, a rental schedule and a rental-income closure.
- **the capacity door's reduced form** — an endogenous utilization externality $A_i^{\text{eff}} = A_i (y_i/\lambda_i)^{-\delta}$ inside the unit-cost hook: demand raises marginal cost, zero profit is preserved, and *no rent is created*, so there is no income-closure problem at all — a one-line change in the cost function with one new parameter. With the opposite sign ($\delta < 0$) it is the Kaldor–Verdoorn case: increasing returns make prices *fall* with demand — variety in the opposite direction. The cheapest door of all.
- **the external door** — endogenous import prices $p_i^M = p_{i0}^M (M_i/M_{i0})^\zeta$ with $\zeta \geq 0$: an upward-sloping foreign supply curve. Import prices enter the intermediate index and the CPI through the margins, so relative prices become functions of import volumes and hence of demand. Moderate surgery, and it exploits precisely the external-account machinery of ADR-0019: the programme's import content bids up import prices — a structuralist absorption / terms-of-trade channel. New candidate.
- **the technology door** — supply shocks $A \neq 1$: this arm, the present workplan; it separates BETA from ALPHA but does not repair the demand-only matrix.

The five doors (added v2). The ex ante derivation now lives in `paper/equivalence.tex`: the zero-profit block is homogeneous of degree one in (p, w) and demand-free (Lemma 1 there), so under $A = 1$ every regime sits at $p = w = \text{CPI} = 1$ and the real wage is pinned at its anchor — the quantity block reduces to one affine multiplier system in which none of $(\theta, \epsilon, \sigma, \eta_s)$ appears. Variety therefore requires demand to enter the price block, and the list above is exactly the door menu: labour, markup, capacity (with its reduced form), external, technology.

Ruled out ex ante (the no-go corollary, Corollary 3 of `paper/equivalence.tex`): no rule that writes the *nominal* wage as a function of endogenous aggregates — an aggregate wage curve $w = \bar{w} (\alpha^\top y / \bar{L})^\varphi$, nominal indexation, or a numeraire change — can create demand-sensitive real outcomes. The price block still forces $p = w \pi(A)$ and $\text{CPI}(p) = w \text{CPI}(\pi)$, so the real wage is invariant to whatever sets w ; in the fixed-wage gauge such a rule is either vacuous or over-determines the system (it forces $\alpha^\top y = \bar{L}$, silently reverting GAMMA to ALPHA). A rule on the *real* wage is the BETA supply equation rearranged and is already in the model, where the demand-only design pins it to the anchor. Quantity-side levers (endogenous saving rates, endogenous export demand) move the multiplier, not the price block, and add no closure variety. Do not schedule this family.

Once prices respond to demand, BETA separates from ALPHA in the demand matrix as well: the numeraire then delivers a real wage that moves with the shock, so $L^s = \bar{L} (w/P)^{\eta_s}$ bites. The sequencing this implies: supply-side arm first (cheap, earns the identification claim for η_s), then sector-specific wages (the repair that gives the published matrix variety), with the markup ladder kept as a robustness arm rather than as the headline mechanism.

Sequencing update (v2). The ranking now reads: the supply-side arm first (unchanged — cheap, earns the identification claim for η_s); then sector-specific wages (the labour door, still candidate 1 for the published matrix — it repairs the BF row's common-wage defect); the utilization/Verdoorn externality as the cheap third arm (no income-closure problem; with $\delta < 0$ it delivers Kaldorian variety in the opposite direction); the terms-of-trade arm as the open-economy candidate with the best fit to the ADR-0019 external account; the markup ladder and the explicit fixed factor kept as robustness arms. The nominal-wage-rule family is ruled out ex ante (see above) and stays off the workplan.

8 Risks

- **Solver.** The pilot used `solve_beta` with eight continuation rungs and passed at $\eta_s = 5$; the `prog` shock at $\alpha = 0.20$ hits the construction block harder, so budget more rungs and keep the warm start.
- **Magnitude.** A uniform shock must stay small (one to five per cent); the single-sector shock can be large because its consumption weight is small.
- **Interpretation.** The arm changes the *experiment*, not the closure; nothing in `src/` moves. It does not fix the BF row (that is candidate 1, sector-specific wages), and it does not make η_s identified without a supply shock — the demand-only matrix stays a null result for this parameter, which is itself the finding.

9 Deliverables

- `experiments/designs/supply_etas_{s1,prog,unif}.toml` with preregistration records.
- One run manifest per cell under `runs/supply_etas-*/`, rows in `registry/scenarios.csv`, index rows in `runs/index.csv`.

- ADR-0018 (design schema for supply shocks) and, if the canary or the round-gain check fails under $A \neq 1$, a follow-up ADR.
- `paper/tables/supply_etas_flows.md`, generated from the manifests and citing the run ids (ADR-0004).

10 Revision Log

- **Version 1** (September 2026)
- **Version 2** (September 2026) — Section 5 extended: the three demand-sensitive-price candidates completed to the five-door menu (new: the external door — endogenous import prices / terms of trade; the capacity door's reduced form — the utilization/Verdoorn externality with no income-closure problem); the nominal-wage-rule family ruled out ex ante (Corollary 3 of `paper/equivalence.tex`); sequencing updated.
- **Version 3** (September 2026) — The schema ADR is drafted as ADR-0023 (proposed); the bridge step is done: probe15 re-measures the pilot on the sectoral kernel (rung separation, the shock-independent nesting, the price-response and welfare-sign findings); the identification-vehicle decision is fixed (scalar single-wage BETA carries the scalar signature, the sectoral uniform vector the per-sector robustness row); workplan step 1 and the canary gate updated.
- **Version 4** (September 2026) — The arm is executed: `supply_etas_s1` 63/63 cells, all gates pass (design commit 202837a; ADR-0023 schema ratified). The preregistered signatures hold: identification exact and magnitude-invariant (implied $\eta_s = \text{input to } 1e-6$); ALPHA control ($L = 1, w = 1.0049$); real GDP monotone in η_s ; F2 = F3 financing neutrality extends to supply shocks; the external-account canary is asserted at $A \neq 1$ on every cell. Flow table `paper/tables/supply_etas_flows.md`; remaining designs `supply_etas_prog` and `supply_etas_unif`; the DOCS_ASSESSMENT section 4.2 write-up is recorded for the manuscript pass.